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Interpretation of the phenolation and structural changes of lignin in a novel ternary deep eutectic solvent

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ABSTRACT

Deep eutectic solvents (DES) are promising green solvents for depolymerization and reconstruction of lignin. Revealing the transformations of lignin during DES treatment is beneficial for high potential lignin applications. In this study, bagasse lignin was treated with a binary DES and three ternary DESs, respectively. The results showed that net hydrogen bonding acidity(α-β) value of DES was positively correlated to the increment of phenolic hydroxyl of lignin, and the ternary DES of choline chloride-formic acid-oxalic acid (ChCl-FA-OA) exhibited the best phenolation performances. The phenolic hydroxyl content of ChCl-FA-OA treated lignin was increased by 50.4 %, reaching 2.41 mmol/g under optimum conditions (120 °C, 4 h, ChCl-FA-OA = 1:2:1). Moreover, it was found that the cleavage of β-O-4′ aryl ether bond and ester bond were dominant reactions during the treatment, accompanied by condensation reactions. Additionally, the obtained lignin oil contained various syringyl and guaiacyl derived phenolic compounds. Especially, the content of acetovanillone in lignin oil reached 29.94 %, much higher than in previous studies. Finally, the degradation mechanism of lignin in ChCl-FA-OA system was proposed. The present work provided insights into the relationship between lignin phenolation and DES properties. The novel ChCl-FA-OA system can achieve efficient lignin depolymerization, and convert lignin biomass into value-added chemical products.

1. Introduction

Biomass is an important substitute for non-renewable fossil materials [1]. With the depletion of fossil fuels and the deterioration of the ecological environment, the utilization of carbon neutral biomass resources is increasingly valued [2]. Lignocellulose is an abundant biomass resource with an annual worldwide production of 2×10^{11} tons according to the report by Francois et al. [3]. It is the most promising raw material for producing biofuels and chemical products because of its low costs and richness. To realize the high-value utilization of lignocellulose, it is necessary to break its anti-degradation barrier to achieve the separation of cellulose, hemicellulose and lignin, and then transform each component to various bio-based materials needed by human beings [4]. Lignin is the second main component of lignocellulose and the most abundant natural aromatic polymer [1,5]. According to a report by Trovagunta et al. [6] in 2022, the annual worldwide production of lignin was 70 million tons. It is a complex three-dimensional amorphous phenolic polymer produced by enzymatic coupling of *p*-hydroxyphenylpropane (H), guaiacyl propane (G), and syringal propane (S), which derive from p-coumarin, juniperol, and sinapinol, respectively [7,8]. Lignin contains a variety of chemical functional groups, including hydroxyl, benzene ring, carboxyl, methoxy groups [5]. Accordingly, it is a potential raw material for preparing renewable aromatic chemicals, biofuels and various lignin-based functional materials, which show broad application prospects in agriculture, chemical and pharmaceutical industry [9–11]. The phenolic hydroxyl group is the most active hydroxyl functional group in lignin, improving the biological activity of lignin. And it has excellent antioxidative, antibacterial, and anti-UV properties [12]. The phenolic modification of lignin can increase the activity of the copolymerization of lignin with phenol and formaldehyde

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and the perform-ance of modified phenolic resin [13]. Currently, the main phenolation methods are based on the reaction between lignin and phenol under strong acids and strong alkaline [14,15]. Nevertheless, this reaction process is poisonous and dangerous. And some side reactions can appear [16]. Accordingly, it is necessary to explore new phenolation methods with mild and sample reaction conditions.

The depolymerization methods of lignin include acid/alkali hydrolysis, pyrolysis, redox, microbial degradation, enzymatic hydrolysis, hydrocracking, metal catalyzed degradation, supercritical fluid or ionic liquid assisted degradation, etc. [17–21]. Disadvantages of main depolymerization methods currently used are obvious due to the complex composition and structural heterogeneity of lignin. They are usually difficult to operate, have high costs, and cause environmental pollution issues, which hinder the high-value utilization and efficient conversion of lignin. In recent years, applying deep eutectic solvents (DES) composed of hydrogen bond acceptors (HBA) and hydrogen bond donors (HBD) in biomass treatment have attracted the interest of many researchers due to its advantages of low cost, low toxicity, biodegradability, easy availability, stability and sustainability [22]. During the pretreatment stage, DES can destroy the compact structure of lignocellulose by competing for hydrogen bonds between lignin and carbohydrates, improving the efficiency of lignocellulose separation and lignin extraction [23]. Moreover, the acid-base catalytic properties of DES enable it to selectively break the ether bonds between phenylpropanes, β-O-4′ bond and C—C bond in the lignin. Accordingly, some researchers also used them for the depolymerization and modification of lignin [24]. Compared with other lignin depolymerization methods, DES depolymerization reaction proceeds under mild conditions. The process is safe, simple and environmentally friendly.

Among various DES systems, DES composed of choline chloride and carboxylic acids, alcohols, amides, etc., has advantages such as low cost, easy preparation, low viscosity, and high biodegradability [25]. ChCl-urea, ChCl-oxalic and ChCl-ethylene glycol show promising potential for biomass pretreatment [26]. It has been reported that acidic DES synthesized using acidic HBD has higher lignin extraction efficiency compared to other types of DES [27]. Tan et al. [28] applied acidic DES synthesized from nine different organic carboxylic acid hydrogen bond donors in the pretreatment of lignocellulosic oil palm empty fruit bunch (EFB) and found that chloride-formic acid (ChCl-FA) had the best extraction rate. Zhang et al. [29] reported that choline chloride-oxalic acid (ChCl-OA) DES achieved the highest delignification rate (98.5 %) compare to DES prepared by ChCl, monocarboxylic acids, dicarboxylic acids, and polyols when treating corncob at 90 °C for 24 h. Hong et al. [30] prepared DES with choline chloride-oxalic acid (ChCl-OA) to treat alkali lignin and successfully obtained S- and G-derived diketones from lignin oil. In addition, researchers also have revealed the structural changes and degradation mechanisms of lignin in binary DES treatment [26,31–33]. For example, Li et al. [26] showed that the slight cleavage of β-O-4′ linkages, preferential breakdown of G units, and demethylation reaction during alkaline ChCl-based DES (ChCl/urea and ChCl/imidazole) treatment, and oxidation only appeared in ChCl/urea system. Hong et al. [33] found the cleavage of β-O-4′ bonds, partial carbon-carbon bonds, dehydration of hydroxyl radicals in lignin's side-chain, and demethoxylation reaction during binary DES (ChCl-FA) treatment. It is known that by adding a third component, the physical and chemical properties of DES can be effectively regulated. The addition of alcohols reduced the density and viscosity of DES, while organic acids exhibited the opposite effects [34]. In this regard, ternary DES are supposed to have better performance than their binary counterparts during the pretreatment process [35]. Researchers found that ternary DES containing two types of HBD can improve lignin removal, enzymatic hydrolysis, and saccharification during lignocellulose pretreatment [22,36]. However, at present, major researches focused on applying ternary DES in the treatment of lignocellulose, few studies were reported on its applications for the degradation process of lignin. Unveiling the structural changes of lignin during ternary DES treatment is beneficial for improving the potential value of lignin in biorefinery processes.

In this study, lignin extracted from bagasse by using formic acid was used as the raw material. Based on the choline chloride-formic acid binary DES (ChCl-FA), three types of ternary DES, including choline chloride-formic acid-ethylene glycol (ChCl-FA-EG), choline chloride-formic acid-urea (ChCl-FA-UR) and choline chloride-formic acid-oxalic acid (ChCl-FA-OA) were prepared by adding a second type of HBD, urea (alkaline), oxalic acid (binary acid and acidic), and ethylene glycol (diol and neutral) to the ChCl-FA binary DES, respectively. All the obtained binary and ternary DES were applied to treat bagasse lignin (BL). The effects of the second HBD on the DES properties were investigated, and their phenolation performances were evaluated. Then the ternary DES with the best phenolation abilities was chosen to further investigate the effect of reaction conditions on the phenolic hydroxyl content and yield of regenerated lignin. The mechanism of lignin degradation by ternary DES was explored by comparing the structure and functional group changes of lignin before and after the treatment. Finally, the products of ternary DES treatment were collected and characterized to determine their compositions. This study aims to provide a guidance for designing optimal ternary DES system for lignin phenolation and degradation, to understand the lignin transformations during ternary DES treatment and promote high-value utilizations of lignin.

2. Results and discussion

2.1. Fourier transform infrared (FT-IR) analysis of lignin

FT-IR analysis was conducted to investigate the functional group changes of BL after DES treatments. The results were shown in Fig. 1. According to previous literatures [37], main characteristics peaks and their corresponding functional groups were listed in Table S1. The ratio of the intensities of A_x/A_{1513} (x = characteristic peaks wavenumber) was used to semiquantitatively analyze the change in the functional group content in the lignin [32], and the results were listed in Table S2. It can be seen that the FT-IR patterns of all regenerated lignin (RL) were similar to the untreated BL, indicating that the basic structure of lignin was not damaged during DES treatment. The peak at 1722 cm^{-1} was attributed to the presence of p-hydroxycinnamate units, such as ferulic acid and p-coumaric acid [38]. The peaks at 1168 cm^{-1} was the characteristics peak of HGS type lignin [37]. Its intensity in $RL_{ChCl\text{-}FA\text{-}OA}$ spectrum was reduced, which was due to the destruction of ester bonds. The reduction of intensity at 835 cm^{-1} may be ascribed to partial degradation and condensation of S-type and H-type lignin. By comparing the calculation results of intensity ratio from Table S2, it can

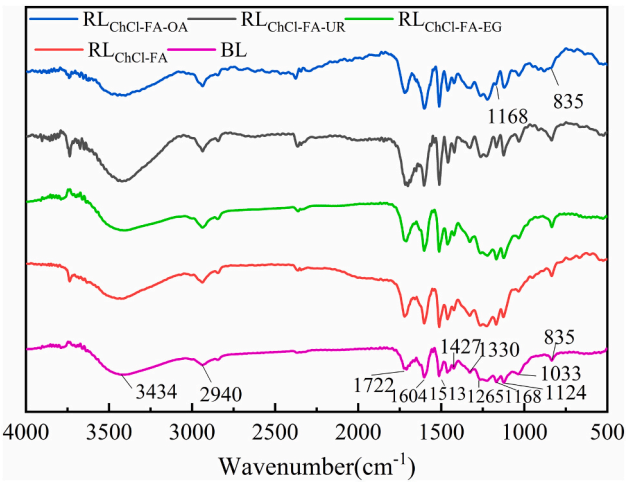


Fig. 1. FT-IR spectra of lignin.

be seen that ChCl-FA-OA exhibited the strongest degradation effect on BL among these four DES systems according to the spectra.

2.2. Effect of the second HBD on the phenolic hydroxyl content and yield

The effects of DES synthesized by different second HBD on the phenolic hydroxyl groups and the yield of RL were investigated. The reaction temperature was 120 °C and the reaction time was 4 h. The results were shown in Fig. 2. It can be seen that the phenolic hydroxyl content of all four RL increased after DES treatment, which was attributed to the cleavage of β-O-4′ aryl ether bonds. The cleavage made phenolic hydroxyl groups more accessible in the lignin, improving its reactivity. Among them, the phenolic hydroxyl content of $RL_{ChCl\text{-}FA\text{-}OA}$ was the highest, reaching to 2.42 mmol/g, implying the addition of oxalic acid contributed to breaking β-O-4′ aryl ether bonds in lignin. Its reason might be that OA provided additional acidic protons. In comparison, the phenolic hydroxyl content of other two ternary DES systems, $RL_{ChCl\text{-}FA\text{-}EG}$ and $RL_{ChCl\text{-}FA\text{-}UR}$, was lower than $RL_{ChCl\text{-}FA}$. The reason might be that the EG in ChCl-FA-EG prevented β-O-4′ aryl ether bonds from being destroyed during degradations [39], and the urea in ChCl-FA-UR reacted with formic acid, reducing the acidity of the solvent, resulting in a decrease in their phenolation efficiency. The yield of RL after DES treatment was between 50 %–72 %, meaning that the degradation rate of DES was relatively low. Higher content of phenolic hydroxyl group in RL means DES possessed higher phenolation efficiency. Therefore, considering ChCl-FA-OA possessed the highest phenolation efficiency, it was chosen to further study the effect of treatment conditions on phenolic hydroxyl content and the structure changes of lignin during the degradation.

2.3. Kamlet-Taft solvatochromic parameters analysis

Kamlet-Taft parameters were determined to quantify the performance of solvents and analyze the effect of solvent hydrogen bonding on lignin phenolation. The parameters α, β and π* represent the solvent hydrogen bond donating ability (hydrogen bond acidity), hydrogen bond accepting ability (hydrogen bond basicity), and polarizability, respectively [40]. The Kamlet-Taft parameters for different DES and the increment of phenolic hydroxyl (ΔPh-OH) were listed in Table 1, fitting curve between the value of α-β (net H-bond donating ability) and the increment of phenolic hydroxyl was shown in Fig. 3. By combining the results of Kamlet-Taft parameters of DES with phenolic hydroxyl content of RL, it can be found that RL treated with DES with higher value of α-β tended to contain higher phenolic hydroxyl content (Ph-OH). Linear fitting between the values of α-β and the increment of phenolic hydroxyl groups (ΔPh-OH) revealed a good positive correlation ($R^2 = 0.96$). The

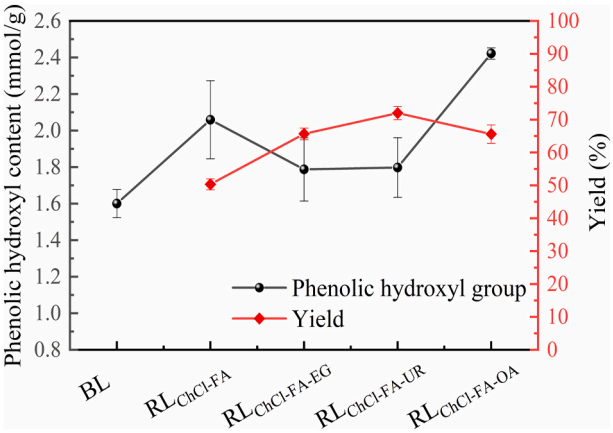


Fig. 2. Effect of the second HBD additive on the content of phenolic hydroxyl groups and yield in RL.

Table 1
Kamlet-Taft parameter values and the increment of phenolic hydroxyl (ΔPh-OH).

parameter	ChCl-FA	ChCl-FA-EG	ChCl-FA-UR	ChCl-FA-OA	ChCl-OA (1:1)
α	1.79	1.33	1.28	1.81	1.80
β	0.02	0.57	0.64	−0.41	0.12
π*	1.38	1.01	0.97	1.40	1.39
α-β	1.77	0.76	0.64	2.23	1.68
Ph-OH	2.059	1.789	1.798	2.423	1.974
Δ(Ph-OH)	0.459	0.189	0.198	0.823	0.374

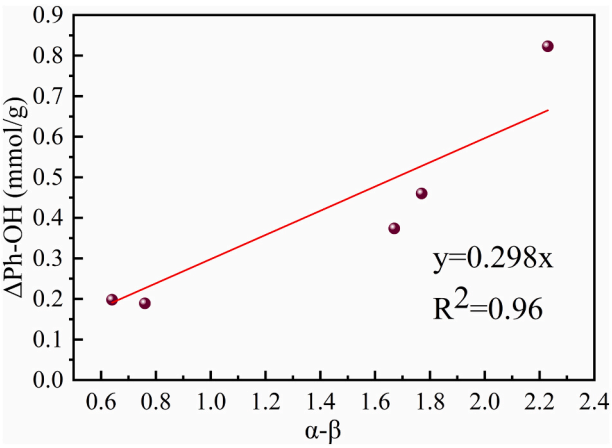


Fig. 3. Fitting curve between α-β value and the increment of phenolic hydroxyl (ΔPh-OH).

value of α-β represented the ability of DES attacking β-O-4′ aryl ether bonds in the lignin. DES with a high value of α-β was prone to form strong hydrogen bonds with lignin and destruct β-O-4′ aryl ether bonds, thereby promoting the lignin degradation [30]. It is also found that the value of β was negatively correlated with the increment of phenolic hydroxyl, and there was no obvious correlation between the values of α and π* and increment of phenolic hydroxyl.

Based on the analysis of Kamlet-Taft parameters, it is expected that DES with high α-β value can be designed to increase the phenolic hydroxyl content of RL and optimize the phenolation process. The results of Kamlet-Taft parameters also confirm that the phenolation abilities of ChCl-FA-OA were superior to the other three DES in our study and suitable for further characterization and evaluation.

2.4. Effect of reaction temperature on the phenolic hydroxyl content and yield of RL

ChCl-FA-OA was chosen as the solvent. Effects of reaction temperature on the phenolic hydroxyl content and the yield of $RL_{ChCl\text{-}FA\text{-}OA}$ was investigated. The reaction time was 4 h, and the molar ratio of ChCl-FA-OA composition was 1:2:1. The results were shown in Fig. 4. It is observed that the phenolic hydroxyl content of $RL_{ChCl\text{-}FA\text{-}OA}$ increased at 120 °C and remained almost unchanged at higher temperatures. This suggests that the DES degradation of lignin was similar to catalytic reactions, most breakages of β-O-4′ aryl ether bond took place within a certain temperature range, leading to the generation of a large amount of phenolic hydroxyl groups [41]. Afterwards, the phenolic hydroxyl content remained at a stable level although the temperature kept increasing. This might be because the simultaneous dehydration and condensation reactions had reached an equivalence [30]. It is also observed that yield of $RL_{ChCl\text{-}FA\text{-}OA}$ decreased with an increase in the reaction temperature. It was due to the degradation of lignin, smaller monomers after the degradation dissolved in water. Accordingly, the

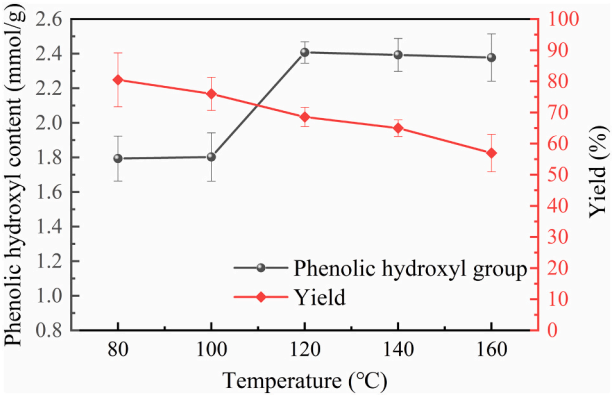


Fig. 4. Effect of temperature on the content of phenolic hydroxyl groups and yield of RL.

optimal reaction temperature for our degradation process was 120 °C.

2.5. Effect of molar compositions of ChCl-FA-OA on the phenolic hydroxyl content and yield of $RL_{ChCl\text{-}FA\text{-}OA}$

The phenolic hydroxyl content and yield were closely related to the composition of DES solvent. ChCl-FA-OA systems with different molar ratios were prepared to investigated the effect of DES molar compositions on the phenolic hydroxyl content and yield of $RL_{ChCl\text{-}FA\text{-}OA}$. The reactions were conducted at 120 °C for 4 h. The results were shown in Fig. 5. With the increase of oxalic acid addition, the phenolic hydroxyl content of $RL_{ChCl\text{-}FA\text{-}OA}$ increased. The $RL_{ChCl\text{-}FA\text{-}OA}$ treated with ChCl-FA-OA system achieved a higher yield than that of the ChCl-FA system. It is worth noting that the phenolic hydroxyl content of RL obtained from the ChCl-OA (1:1) treatment was equivalent to that of the ChCl-FA (1:2) treatment, but both of them were lower than the ChCl-FA-OA (1:2:1) DES treatments. This means that the ChCl-FA-OA ternary DES systems had better treatment capabilities than their corresponding binary DES systems, which might be attributed to the synergetic effect of FA and OA during the degradation. When the molar ratio of ChCl-FA-OA composition was 1:2:1, the phenolic hydroxyl content was the highest. Since the viscosity of OA was obviously higher than that of FA, further addition of OA cannot obtain a clear and uniform solution at room temperature. In this regard, the optimal molar ratio of ChCl-FA-OA was 1:2:1 for our study.

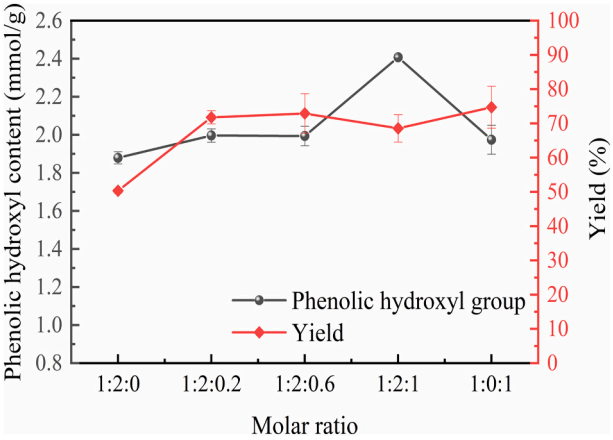


Fig. 5. Effect of solvent molar ratio on phenolic hydroxyl content and yield of RL.

2.6. Effect of reaction time on the phenolic hydroxyl content and yield of $RL_{ChCl\text{-}FA\text{-}OA}$

Effects of reaction time of ChCl-FA-OA treatment on the phenolic hydroxyl content and yield of $RL_{ChCl\text{-}FA\text{-}OA}$ were investigated with a molar ratio of 1:2:1 at 120 °C. As shown in Fig. 6, the content of phenolic hydroxyl groups firstly increased and then decreased with the extension of reaction time. The pattern for the relation between reaction time and yield was not that obvious, but it roughly showed the opposite trend. This phenomenon can be explained by the competition between the depolymerization and partial condensation that simultaneously took place during the lignin degradation. The depolymerization was the dominant reaction at the beginning, the breakage of ether bonds increased the phenolic hydroxyl content and generated small molecules, leading to a yield reduction. After reaching the highest content of phenolic hydroxyl groups at a certain time, the impact of condensation reaction gradually became stronger, more and more small molecules were aggregated and condensed into insoluble large molecules, which decreased the phenolic hydroxyl content while increased the yield of $RL_{ChCl\text{-}FA\text{-}OA}$. Since the phenolic hydroxyl content of $RL_{ChCl\text{-}FA\text{-}OA}$ reached the highest value (2.41 mmol/g) at 4 h, the optimal reaction time for our process was 4 h.

2.7. ^1H NMR analysis

^1H NMR analysis was used to identify the structure of lignin. The ^1H NMR spectra of BL and RL were shown in Fig. 7. The peak signal assignments were listed in Table S3 according to previous literatures [42,43]. The signal between 8.0 and 6.2 ppm corresponded to the aromatic proton in the lignin aromatic ring region [44]. The signal between 7.75 and 7.3 belonged to the hydrogen atom on the aromatic ring of *p*-hydroxyphenyl (H) unit and p-coumaric acid and ferulic acid. Its intensity was considerably decreased in the spectrum of $RL_{ChCl\text{-}FA\text{-}OA}$, suggesting that the structure of para coumaric acid ester or the *p*-hydroxyphenyl (H) unit was degraded, while the signal between 7.75 and 7.3 had hardly changed in the spectrum of $RL_{ChCl\text{-}FA}$. It confirmed the depolymerization effect of ChCl-FA-OA was better than that of ChCl-FA. The signal at 5.0 to 4.1 ppm was the H signal in the β-O-4′ structure, its intensity also obviously declined in the RL, implying that the β-O-4′ bond was broken. These results proved that depolymerization was promoted by the DES treatment during the lignin degradation. Moreover, the signal intensity at 3.73 ppm clearly decreased in the RL, which suggested demethoxylation reactions also occurred during the degradation process. Demethoxylation can produce more reactive sites toward the preparation of lignin-based phenolic resins [45]. A new small peak appeared at 3.41 ppm was the H signal from ChCl in DES (confirmed by Fig.S4), which might be due to a slight reaction between

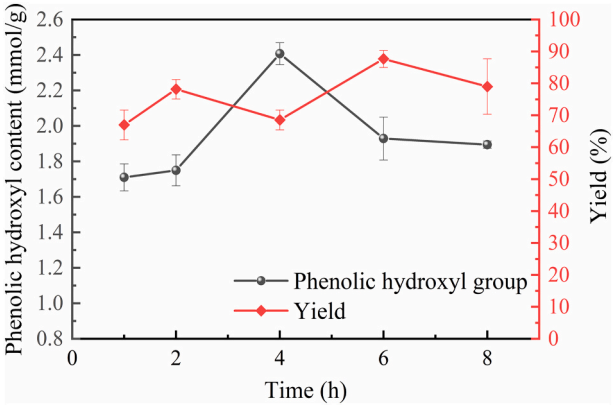


Fig. 6. Effect of reaction time on the content of phenolic hydroxyl groups and yield in RL

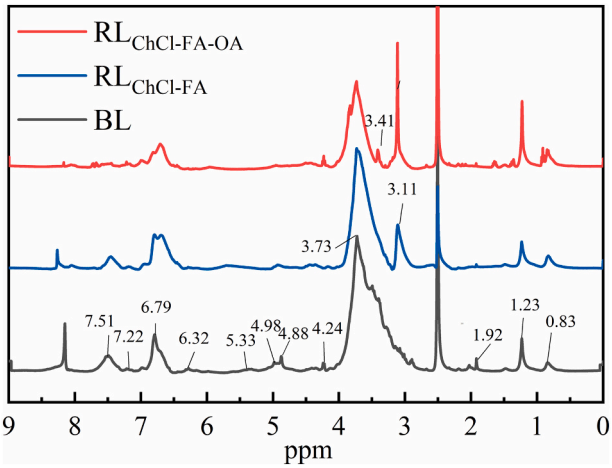


Fig. 7. ^{1}H-NMR spectra of lignin.

lignin and ChCl during the degradation process and a small amount of residual ChCl in RL [32]. The signal peak at 3.11 ppm might be the residual DES (confirmed by Fig. S4 and Fig. 8) [46]. The signal at 1.23 ppm represented H of the aliphatic group. The reason that signal intensity of $RL_{ChCl\text{-}FA\text{-}OA}$ was significantly increased might be the formation of condensed interunit C—C bonds resulting in more aliphatic groups [47].

2.8. ^{13}C NMR analysis

^{13}C NMR spectroscopy was applied to quantify the side chain linkages and elucidate the condensation of lignin fractions. The results were shown in Figs. S1, S2 and S3. The quantitative analysis of ^{13}C NMR data was carried out based on the signals in the spectral region between 162.0 and 102.0 ppm. The content of each structural units was calculated according to the method described in previous literatures [48,49]. The aromatic ring area was divided into oxidation area (C—O, 161.0–141.0 ppm), condensed aromatic ring area (C—C, 141.0–125.0 ppm) and protonation aromatic ring area (C—H, 125.0–102.0 ppm), respectively. It was found that after DES treatment, the number of C—C aromatic hydrocarbons in RL increased (BL = 1.79/ Ar, $RL_{ChCl\text{-}FA}$ = 1.96/ Ar, $RL_{ChCl\text{-}FA\text{-}OA}$ = 2.07/ Ar). This confirmed the occurrence of condensation reaction during the degradation process. Notably, the -CH_3O content decreased (BL = 1.79/ Ar, $RL_{ChCl\text{-}FA}$ = 1.30/ Ar, $RL_{ChCl\text{-}FA\text{-}OA}$ = 1.54/ Ar), demonstrating that demethoxylation also took place during the treatment. In addition, the signal reduction between 102 and 90 pm was associated with the removal of carbohydrates from RL, and the absence of peak signals between 90 and 72 ppm was due to the destruction of ether bonds during the treatment [24].

2.9. 2D HSQC-NMR analysis

2D HSQC-NMR can determine the specific carbon-hydrogen functionalities to analyze changes of inter-unit linkages in lignin that are unable to be identified by ^{1}H NMR spectra and ^{13}C NMR spectra. The HSQC spectra were depicted in Fig. 8. The results of semiquantitative analysis were listed in Table S4. According to previous literatures [31,33,50,51], the main signal assignments in the spectra were listed in Table S5. As shown in Fig. 8 and Table S4, in the side-chain (δ_C/δ_H 50–90/2.5–6.0), compared with BL spectrum, the signal for A_γ' substructure in $RL_{ChCl\text{-}FA\text{-}OA}$ spectrum was enhanced, suggesting that the acylation reaction occurred between the γ-OH and COOH group in DES. A_γ' substructure in $RL_{ChCl\text{-}FA}$ disappeared, which was possibly due to the deacylation reaction at high temperatures [33]. Additionally, loss of C_γ signal of C substructures in $RL_{ChCl\text{-}FA\text{-}OA}$ spectrum indicated that C—C bonds (β-β') were cleaved. The Hibbert' ketone signal was detected in the $RL_{ChCl\text{-}FA\text{-}OA}$ spectrum, verifying the effect of ChCl-FA-OA on the cleavage of β-O-4′ bonds, and the degradation process of lignin in ChCl-FA-OA was acid catalyzed reactions [24,31]. Moreover, the X signals in RL spectrum were not detected, which was due to the removal of carbohydrates. Accordingly, the purity of RL was increased. The purity of $RL_{ChCl\text{-}FA\text{-}OA}$ reached 80.3 % by the determination, which is higher than BL (73.8 %).

In the lignin aromatic ring region (δ_C/δ_H 90–150/5.5–8.5) G type, S type, and H type lignin were observed, implying that BL is HGS type lignin. The signal corresponding to para coumaric acid ester ($PCE_{2,6}$) structure was clearly observed in the untreated lignin spectrum. It became weaker in the $RL_{ChCl\text{-}FA\text{-}OA}$ spectrum, verifying the destruction of para coumaric acid ester structure in lignin caused by the ChCl-FA-OA treatment. The reduction of signal for para coumaric acid ester structure was not evident in the $RL_{ChCl\text{-}FA}$ spectrum, verifying that ChCl-FA lacked of ability to break the para coumaric acid ester structure. The oxidation unit appeared in the S unit of $RL_{ChCl\text{-}FA\text{-}OA}$, which may be related to the cleavage of the aryl ether bond in the S unit. In addition, it was found that S-type units in BL were prone to be degraded or removed during DES treatment according to Table S4. These results reaffirmed the occurrence of depolymerization during DES treatment, and the depolymerization effect of ternary ChCl-FA-OA was superior to that of binary ChCl-FA. Moreover, the spectra also provide evidence for the presence of condensation reactions. The chemical shifts of $S_{2,6}$ signals in the $RL_{ChCl\text{-}FA\text{-}OA}$ spectrum were altered, which was due to the formation of condensed C—C linkages in S units [30,52]. The enhancement of signal for H unit was discovered in the RL spectrum, which was attributed to the demethoxylation of G and S unit.

2.10. Analysis of molecular weight and polydispersity index of lignin

The above NMR analysis suggested that both depolymerization and condensation took place during the lignin degradation. Comparison of molecular weights of treated and untreated lignin can further clarify the role of these two reactions. The mass distribution curves of $RL_{ChCl\text{-}FA\text{-}OA}$ and BL from GPC were shown in Fig. 9. The values of molecular weight (M_w and M_n) and polydispersity of lignin were calculated from GPC curves and listed in Table 2. As can be seen, compared with milled wood lignin (MWL) from sugarcane bagasse [53], which possessed similar molecular weight to native sugarcane bagasse lignin, both BL and $RL_{ChCl\text{-}FA\text{-}OA}$ exhibited low weight-average molecular weights, 1205 $g{\cdot}mol^{-1}$ and 1092 $g{\cdot}mol^{-1}$, respectively, suggesting that formic acid extracted low molecular weight lignin from bagasse in the pretreatment. Furthermore, the molecular weight of $RL_{ChCl\text{-}FA\text{-}OA}$ was lower than that of BL, which was ascribed to the depolymerization of BL during ChCl-FA-OA treatment. Therefore, it is reasonable to deduce that the depolymerization was the main reaction in our ChCl-FA-OA treatment, and the effect of condensation was secondary. Moreover, it was found that ChCl-FA-OA treatment was beneficial for increasing the uniformity of lignin molecular weight, the curve of molecular weight distribution of $RL_{ChCl\text{-}FA\text{-}OA}$ was narrower than that of untreated lignin, and the PDI value decreased from 2.41 to 1.86 after the treatment.

2.11. Thermogravimetric analysis of lignin

The thermal stability of lignin was evaluated through thermogravimetric analysis, the results were described in the form of weight loss and derivative weigh against temperature. It can be seen from Fig. 10 that untreated BL and $RL_{ChCl\text{-}FA\text{-}OA}$ exhibited similar thermal decomposition behaviours, and the highest decomposition rate for both of them appeared at 360 °C. This indicates that the thermal stability of lignin remained unchanged after ChCl-FA-OA treatment. The weight loss below 100 °C was mainly due to the evaporation of water. Lignin decompositions mostly occurred at 150–500 °C. The temperature ranging from 230 to 500 °C was mainly for lignin pyrolysis. The first stage pyrolysis was the decomposition of low-molecule-weight fractions in

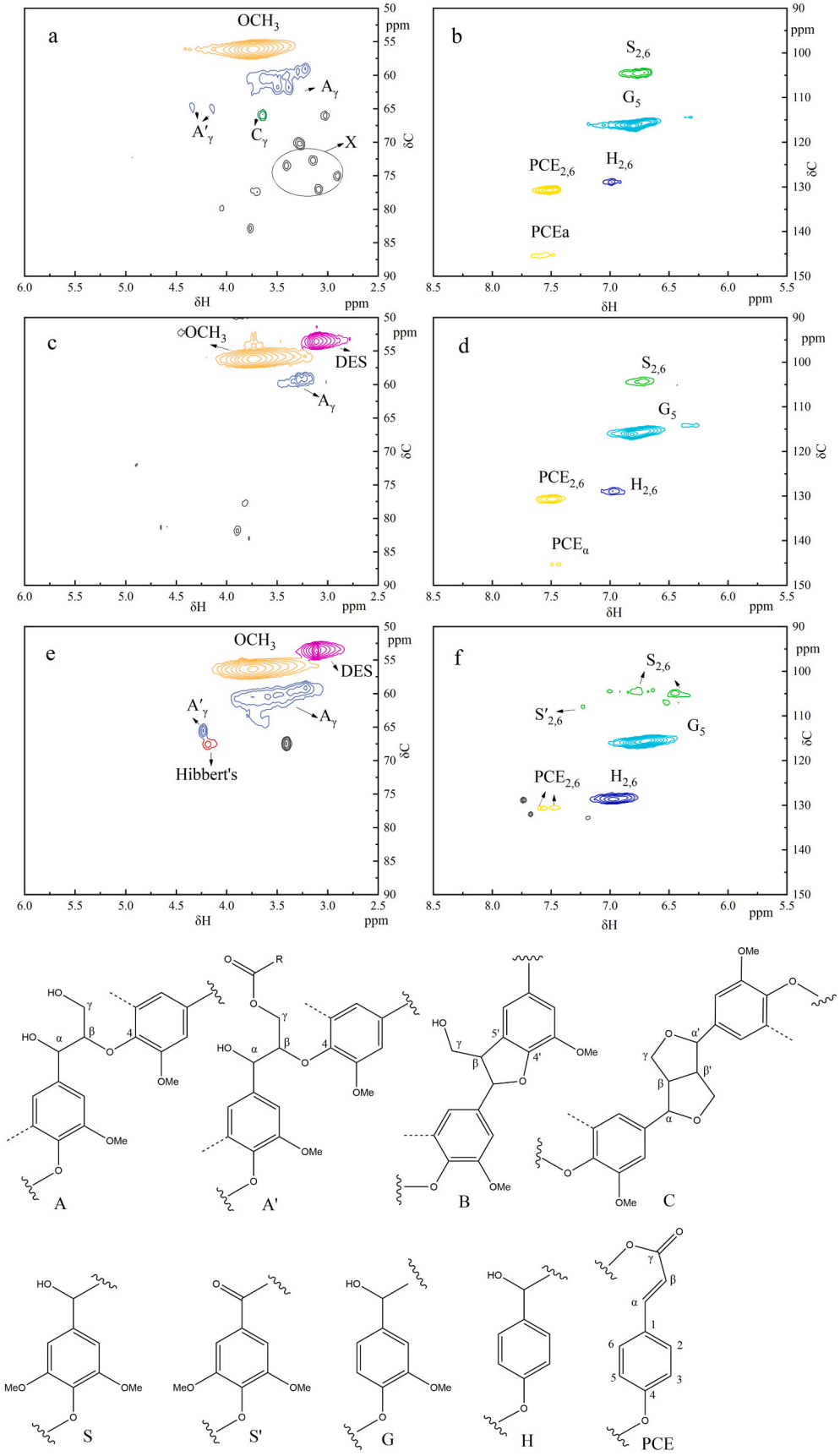


Fig. 8. 2D HSQC-NMR spectra of lignin (a: BL side chain region, b: BL aromatic region,c: $RL_{ChCl\text{-}FA}$ side chain region, d: $RL_{ChCl\text{-}FA}$ aromatic region, e: $RL_{ChCl\text{-}FA\text{-}OA}$ side chain region, f: $RL_{ChCl\text{-}FA\text{-}OA}$ aromatic region.

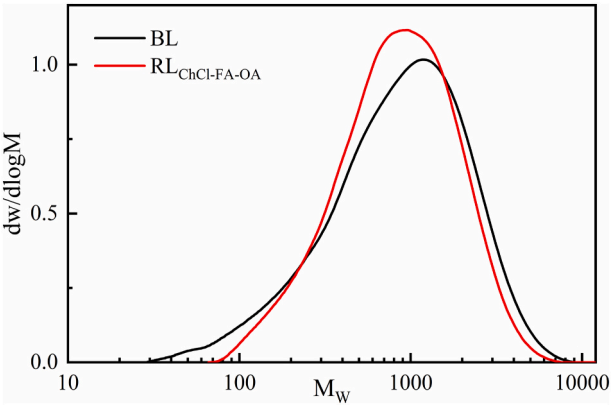


Fig. 9. Mass distribution curve of lignin.

Table 2
Molecular weight and dispersion coefficient of lignin (Mn: Number-average molecular weight; Mw: Weight-average molecular weight).

Samples	Mn	Mw	PDI(Mw/Mn)
MWL	5700	9560	1.68
BL	499	1205	2.41
RL$_{ChCl-FA-OA}$	588	1092	1.86

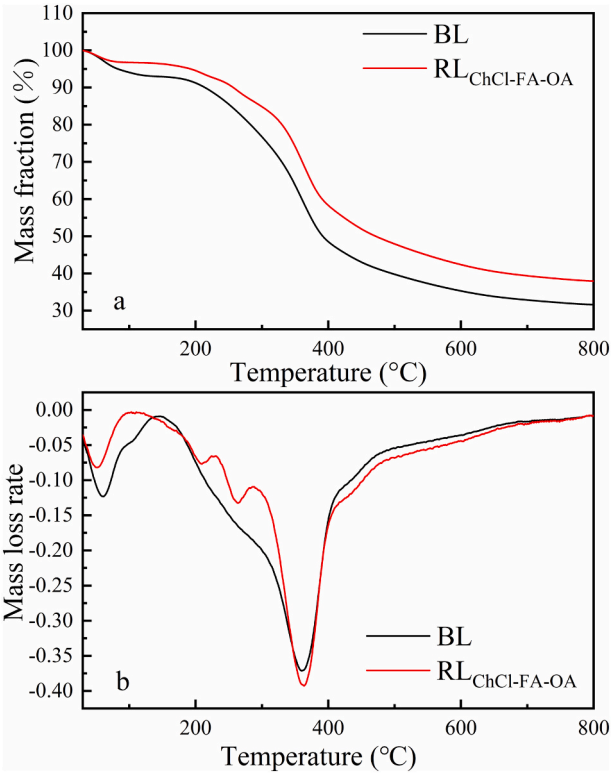


Fig. 10. TG (a) and DTG curves (b) of lignin.

lignin. Since untreated BL contained relatively high content of low-molecular-weight fractions, the weight loss of it was faster than RL at this stage. Then, after 230 °C it was the decomposition of lignin phenylpropane polymer, including side chain oxidation and dehydration, C—C bond cleavage, degradation of aromatic rings, and removal of methoxy groups from aromatic rings [54]. After reaching 500 °C, the decomposition tended to be slow down and the lignin gradually carbonized. After the decomposition process completed, the residual char of RL$_{ChCl-FA-OA}$ (38.9 %) was slightly higher than untreated BL (31.5 %).

2.12. Compositions of lignin oil

To explore the utilization of depolymerization products after lignin degradation and the mechanism of depolymerization, GC–MS was used to identify the lignin oil composition after ChCl-FA-OA treatment. The relevant products were determined according to NIST11 and previous studies [30,31,33], as shown in Fig. 11. It is found that major depolymerization products of lignin were syringyl and guaiacyl derived phenolic compounds, successively including vanillin, guaiacyl propylene, guaiacylacetone, 4-ethyl guaiacol, acetovanillone, syringylacetone, and syringyldiketone. This is consistent with previous findings [30,31,33]. And these products are high-value chemicals that have been widely used in many facets of chemical industry. Moreover, the obtained lignin oils were rich in phenolic diketones, confirming that both ether bond cleavage and oxidation reactions occurred during ChCl-FA-OA treatment [33]. Hong et al. [30] also observed phenolic diketones in their DES pretreatment products. It is worth noting that our products not only contained syringyl derived phenolic diketones (16.34 %), but also relatively high percentage of guaiacyl derived phenolic acetovanillone (29.94 %). To the best of our knowledge, there are only a few reports about obtaining acetovanillone by DES degradation, and the percentage of acetovanillone in products was usually very low [30–33]. Acetovanillone exhibits high potentials in advanced pharmaceutical applications. It can be used as an effective inhibitor of NADPH oxidase and enhance antioxidant abilities in different disease models [55–57].

2.13. Mechanisms of lignin degradation in ChCl-FA-OA

Understanding the mechanisms involved in lignin degradation in DES systems is critical to design solvents with tailored properties and enhance understanding of lignin transformation. Combining all the data herein, the cleavage of β-O-4' aryl ether bonds through acid catalysis is the main reaction during the ChCl-FA-OA treatment process. It is attributed to the synergistic effects among the individual components in the solvent. Chloride anions in ChCl form strong H-bonds with hydroxyl groups of lignin, which facilitates the breakage of intramolecular H-bond network of lignin [25]. Acidic protons of HBD (FA and OA) and acylation between the γ-OH group and -COOH (FA and OA) favor the breakage of ether bonds [30]. Notably, the role of OA and FA is slightly different in the DES system. Since the acid strength of OA ($pKa_1 = 1.25$, $pKa_2 = 4.14$) is higher than that of FA ($pKa = 3.75$), OA tends to provide more acidic protons at the same amount usages of acids during the treatment. And besides enhancing the acidity of the system, another

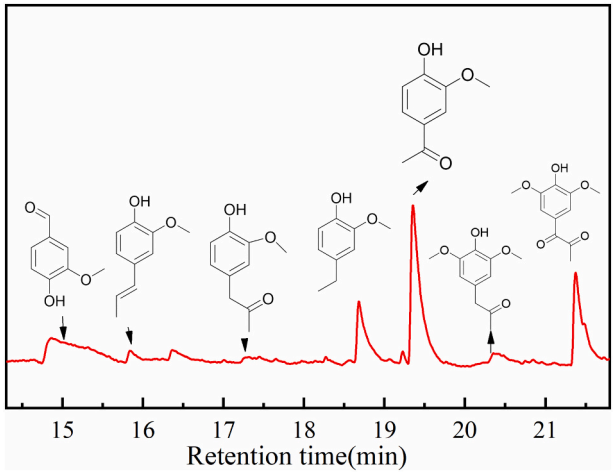


Fig. 11. GC-MS diagram of lignin oil.

important function of FA is to reduce viscosity and regulate solvent fluidity to guarantee a high solubility of lignin required for achieving optimal degradation performances.

The possible reaction pathways of lignin degradation in ChCl-FA-OA were summarized and illustrated in Fig. 12. Firstly, the proton in the solvent attacks α-OH in β-O-4′ structure and releases a molecule H_2O to form carbocation structure A [25]. It is a prevalent route for lignin depolymerization. The elimination reaction between α positive charge and β-H of A produces enol ether intermediate B, and regenerates hydrogen ions at the same time. Then, the hydrolysis of C═C leads to the cleavage of the C—O bond, resulting in the formation of Hibbert's ketone C. The presence of Hibbert's ketone signal was confirmed by the analysis of 2D HSQC-NMR spectra of lignin in Fig. 8. The rearrangement of C generates D, which forms E by isomerization [58]. Finally, oxidoreduction of D and E results in monoketones and diketones compounds, which was consistent with the findings of lignin oil compositions by GC–MS analysis. A similar pathway of DES degradation for lignin was also observed by Hong et al.in their study of binary DES systems [30]. On the other hand, the elimination reaction of A also produces α-β unsaturated compounds F [59], which is depolymerized into dimer G. Then G results in the formation of acetovanillone and 4-ethyl guaiacol. GC–MS analysis of lignin oil revealed that the degradation products contained acetovanillone and 4-ethyl guaiacol. Moreover, from 2D HSQC-NMR analysis, it is demonstrated that γ-OH of β-O-4′ structure reacts with -COOH group of DES, and A condenses with S-type lignin. Furthermore, ^{1}H-NMR and 2D HSQC-NMR analysis showed that para coumaric acid ester ($PCE_{2,6}$) signal weakened, suggesting the cleavage of ester linkage.

2.14. Recyclability of ChCl-FA-OA

Recyclability is significant in evaluating the performances of green solvents. The changes of ChCl-FA-OA before and after treatment were investigated by ^{1}H-NMR and 2D HSQC-NMR spectra. The results were shown in Fig. S5 and S6. The signals of ChCl-FA-OA before and after treatment were similar, suggesting that the chemical structure of the DES was not destroyed during the treatment, and its degradation ability was expected to be maintained at a high level. The weakening of the signal at 8.13 ppm was attributed to the consumption of ChCl-FA-OA.

The phenolation performance of ChCl-FA-OA and the yield of $RL_{ChCl\text{-}FA\text{-}OA}$ were investigated after ChCl-FA-OA was recycled up to three times. The recovery rate of ChCl-FA-OA was between 78 % and 85 % each time. It can be seen from Fig. 13 that the phenolic hydroxyl content of $RL_{ChCl\text{-}FA\text{-}OA}$ decreased from 2.32 mmol/g to 1.92 mmol/g after the first cycle. This reduction of phenolation ability might be due to the consumption of its protons (observed in Fig. S6). The phenolic hydroxyl contents of $RL_{ChCl\text{-}FA\text{-}OA}$ of the second and the third cycle were 1.81 mmol/g and 1.83 mmol/g, respectively, close to the content of the first cycle. It is worth noting that after ChCl-FA-OA was recycled three times, the phenolic hydroxyl content of $RL_{ChCl\text{-}FA\text{-}OA}$ was still higher than that of untreated BL (1.60 mmol/g). Moreover, the yield of $RL_{ChCl\text{-}FA\text{-}OA}$ slightly increased from 72 % to 76 %, suggesting ChCl-FA-OA maintained a relatively stable degradation performance after recycled.

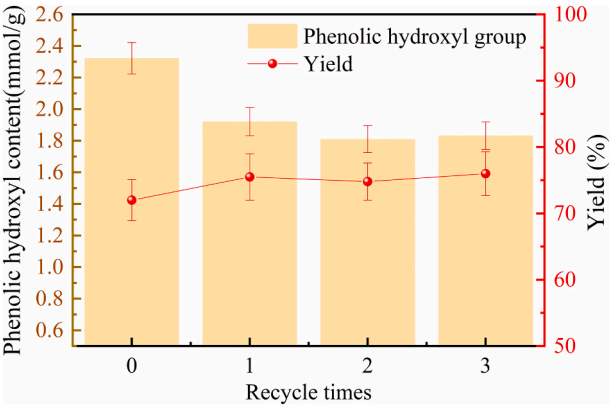


Fig. 13. Effect of recycling times of ChCl-FA-OA on the phenolic hydroxyl content and yield of $RL_{ChCl\text{-}FA\text{-}OA}$.

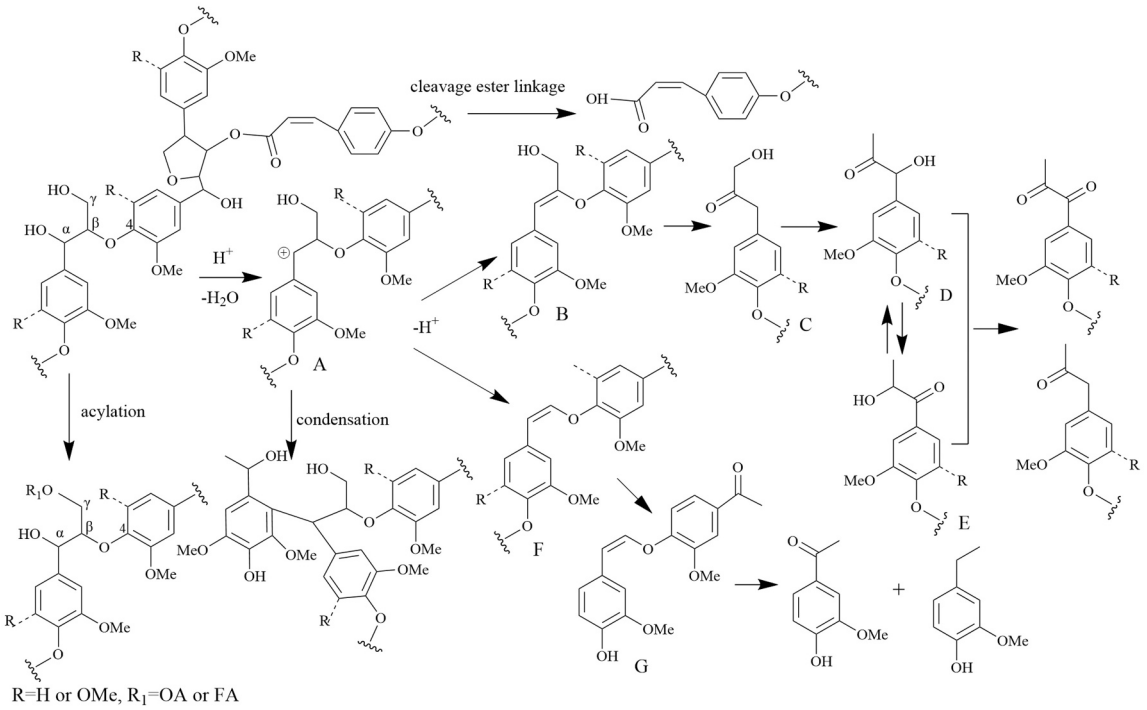


Fig. 12. Mechanism of lignin degradation during ChCl-FA-OA treatment.

3. Conclusions

In this study, BL was treated with a novel ternary DES system. The phenolation ability of DES, and the structural changes and degradation mechanisms of lignin during the DES treatment were explored. The results showed that the α-β value of DES was positively correlated to the increment of phenolic hydroxyl ($R^2 = 0.96$), ChCl-FA-OA had the best phenolation modification efficiency among three ternary DES. ChCl-FA-OA treatment increased the phenolic hydroxyl content of lignin by 50.4 %, reaching 2.41 mmol/g under optimum conditions (120 °C, 4 h, ChCl-FA-OA = 1:2:1), and $RL_{ChCl\text{-}FA\text{-}OA}$ had a lower molecular weight (1092 $g{\cdot}mol^{-1}$), and exhibited a more uniform distribution (PDI = 1.86). The reactivity of lignin was greatly improved. The cracking of the β-O-4′ aryl ether bonds and para coumaric acid ester bond was the dominant reaction in ternary ChCl-FA-OA, while binary ChCl-FA can hardly break para coumaric acid ester structure. The cleavages of β-O-4′ bond resulted from acid catalysis have two possible routes. One is α-carbocation generates α-β unsaturated compounds, the other is elimination reaction produces enol ether intermediates. It is worth mentioning that a large number of phenolic monoketones and diketones were extracted from the lignin oil, especially the content of acetovanillone reached 29.94 %, much higher than that of previous studies. This work revealed the relationship between the DES properties and its phenolation ability, providing a comprehensive understanding of lignin structural changes in ternary DES degradations. The novel ChCl-FA-OA system can be applied to efficient depolymerization and phenolation of lignin, and transform lignin biomass into valuable products applied in the chemical and pharmaceutical industry.

4. Materials and methods

4.1. Materials

The raw material sugarcane bagasse is purchased from Guangxi Nongken Sugar Industry Group Jinguang Sugar Co., Ltd., (Guangxi, China). After washing, it was dried to constant weight in an oven. According to the National Renewable Energy Laboratory (NREL) method for determining the composition of lignocellulose, the main compositions of the sugarcane bagasse material were 42.1 % cellulose, 23.5 % xylan, 1.5 % arabinan, 24.7 % acid-insoluble lignin, and 1.0 % acid-soluble lignin.

Choline chloride and Sodium carbonate decahydrate were purchased from Aladdin Biochemical Technology Co, Ltd., (Shanghai, China). Nile red and p-nitroaniline were obtained from McLean Biochemical Technology Co, Ltd., (Shanghai, China). Ethyl acetate, formic acid (88 %), sodium hydroxide, and ethylene glycol were purchased from Guangdong Guanghua Technology Co., Ltd., (Guangdong, China). Oxalic acid was obtained from Chengdu Aikeda Chemical Reagent Co., Ltd. , (Sichuan , China). Formic acid (98 %) was purchased from China National Pharmaceutical Group Chemical Reagent Co., Ltd., (Shanghai, China). Urea was obtained from Xilong Science Co., Ltd., (Guangdong, China). All the reagents above are analytically pure. Guaiacol, Folin reagent was purchased from Beijing Solarbio Science & Technology Co., Ltd., (Beijing, China).

4.2. Preparation of binary DES

The hydrogen bond receptor choline chloride (ChCl) and hydrogen bond donor formic acid (FA, 98 %) were added to a circular bottom flask, and mixed in a molar ratio of 1:2. The solution was stirred for 1 h in an oil bath at 80 °C until a uniform and transparent liquid was formed, which was denoted as ChCl-FA.

4.3. Preparation of ternary DES

Ethylene glycol (EG), urea (UR), and oxalic acid (OA) were used as the second hydrogen bond donor (x), respectively. With a molar ratio of n (ChCl): n (FA): n (x) = 1:2:1, the second hydrogen bond donor was mixed with ChCl and FA to prepare a ternary DES. The process was similar to the preparation of binary DES described above. The obtained ternary DES were named ChCl-FA-X.

4.4. Preparation of BL

10.0 g of dried sugarcane bagasse and FA (88 %) were mixed at a solid-liquid ratio of 1:10 (g: mL) in a 500 mL three necked flask. The mixture was stirred at 350 r/min and reacted at 95 °C for 100 min. After the reaction was completed, the mixture was cooled to room temperature for filtration. The filtrate was collected and distilled in a vacuum rotary evaporator at 77 °C to recover the FA solution. The distilled solution was put into a beaker containing deionized water in order to precipitate the lignin, then it was centrifuged to collect solid lignin. Solid lignin was washed four times, and freeze-dried to obtain BL. The purity of BL reached 73.8 % and the yield of BL is about 80–85 %.

4.5. Experiment of DES depolymerize lignin

BL (0.30 g) and DES in a solid-liquid ratio of 1:20 (g: g) were mixed in a three necked flask. Then, the three necked flask containing the mixtures was placed in a constant temperature oil bath at 120 °C and stirred for 4 h. After the reaction was completed and cooled to room temperature, the mixture was slowly added to a beaker containing deionized water, and then centrifuged at 8500 rpm for 10 min to obtain regenerated lignin (RL) and liquid mixtures. The RL sample was centrifuged with water three times and freeze-dried for 48 h. Afterwards, the RL sample was labelled as RL_{DES} based on DES name. On the other hand, liquid mixture was extracted by ethyl acetate. The upper liquid phase was distilled by a rotary evaporator to obtain lignin oil. The lower liquid phase was evaporated to remove water and recover DES at 72 °C.

4.6. Determination of Kamlet-Taft solvatochromic parameters

Two types of dyes, Nile Red (NR) and p-nitroaniline (NH_2), were mixed with methanol to prepare the solution (0.001 mol/L), respectively. Then, 0.5 mL of the solution was added to a 10 mL centrifuge tube and vacuum-dried at 40 °C for 48 h to evaporate the methanol. Afterwards, DES was introduced and the mixture was homogenized. The absorption spectrum of the mixture was measured using a UV-1800 Ultraviolet-visible Spectrophotometer (Shimadzu Corporation, Japan) at room temperature. The background was the spectrum measured from deionized water. All spectra were recorded between 300 and 700 nm. The Kamlet Taft parameters were calculated according to following equations [60].

$$\pi^* = (19.839 - V_{NR})/2.9912 \quad (1)$$

$$\alpha = (19.9657 - 1.0241 \times \pi^* - V_{NR})/1.6078 \quad (2)$$

$$\beta = 11.134\text{–}3580/\lambda(NH_2)_{max} - 1.125 \times \pi^* \quad (3)$$

where α is the solvent hydrogen bonding acidity, β is the solvent hydrogen bonding alkalinity, π* is the solvent polarity, λ_{max} is the maximum absorption wavelength the dye used, and $V = 1/(\lambda_{max} \times 10^{-4})$.

4.7. FT-IR analysis

Infrared spectra of lignin were measured using a Fourier transform infrared spectrometer (MAGNA-IR550, Thermo Corporation, USA). All spectra were collected at a wavelength range of 4000–500 cm^{-1}, with a resolution of 4 cm^{-1} and 50 scans.

4.8. NMR analysis

^{1}H-NMR spectra, ^{13}C NMR spectra, and 2D-HSQC NMR analyses were carried out by a nuclear magnetic resonance spectrometer (AVANCE-III HD 400, Brooke Company, Germany). 80 mg of lignin sample were dissolved in DMSO-d_6 (0.6 ml). ^{1}H-NMR spectrum was registered with relaxation delay of 1.5 s and number scans of 16. ^{13}C NMR spectra were recorded with 10 μs pulse width and number scans of 5120. 2D-HSQC NMR were recorded with the hsqcedetgpsisp sequence, the spectral width of 190 ppm in F1 and number scans of 16.

The 2D-HSQC NMR relative quantitative analysis was performed by the MestReNova software to integrate each inter-unit linkage. Since G_2 unit signals were not detected, the G_5 unit had strong signals before and after processing, the signal intensity of $0.5S_{2,6} + G_5 + 0.5H_{2,6}$ was as an aromatic unit (Ar). Abundances of different inter-unit linkages(X) were expressed as per aromatic unit (X/Ar).

4.9. Thermogravimetric analysis (TGA)

A synchronous thermal analyzer (Mettler TGA/DSC 3+, Switzerland) was used to evaluate the thermal stability of lignin. The tests were conducted under nitrogen environment. The temperature range was 30 to 800 °C, and the heating rate was 10 °C/min.

4.10. GPC analysis

The relative molecular weight of lignin was determined by a gel permeation chromatography (Agilent GPC 50, USA). The mobile phase was tetrahydrofuran (THF).

4.11. Phenol hydroxyl determination and yield calculation

The Folin phenol reagent method was used to determine the phenolic hydroxyl content. Firstly, 0.10 g/mL NaOH solution and 15%wt $Na_2CO_3{\cdot}10H_2O$ solution were prepared for standby. Then 0.010 g lignin sample and 0.2 mL NaOH solution were mixed in a 100 mL volumetric flask and then water was added to tick mark of volumetric flask. Secondly, 1 mL lignin solution, 0.5 mL deionized water and 0.5 mL Folin phenol reagent were successively added to a 10 mL centrifuge tube, and 2 mL $Na_2CO_3{\cdot}10H_2O$ solution was added after 2– 3 min. Finally, the absorbance of mixed solution was recorded at 760 nm wavelength by a UV-1800 Ultraviolet-visible Spectrophotometer (Shimadzu Corporation, Japan) after 40– 60 min. $Na_2CO_3{\cdot}10H_2O$ solution was used as the background. The standard curve of guaiacol between concentration and absorbance was prepared. The phenolic hydroxyl content of lignin was calculated based on the standard curve, and the yield was determined according to Eq. (4).

$$A = m/m_0 \tag{4}$$

where A is the yield; m is the weight of RL, m_0 is the quality of BL.

4.12. Determination of lignin content

The content of lignin was determined according to the procedure described in a previous literature with some modifications [61]. 0.1 g (m_0) of lignin sample and 3 mL of 72%wt H_2SO_4 were added into a 100 mL conical flask. Then the mixtures were hydrolyzed for 2 h at room temperature. Afterwards, 84 mL of deionized water was added into the conical flask, and the conical flask was placed and autoclaved in a high-pressure sterilization pot at 121 °C for 1 h. After completion of the autoclave cycle, the hydrolyzates were slowly cooled to the room temperature. The supernatant of the hydrolysate suspension was then filtered by a 0.22 μm-pore-size filtration system. Absorbance of the filtrate obtained was recorded by a UV-1800 Ultraviolet-visible Spectrophotometer (Shimadzu Corporation, Japan). The sample was diluted to ensure its absorbance at 205 nm wavelength was between 0.2 and 0.7. 4%wt H_2SO_4 was used as the reagent blank. The remaining hydrolysates were filtered through a Buchner funnel, then the filter residue and the Buchner funnel were dried at 105 °C for 24 h. Total weight of filter residues and Buchner funnel before and after the dry process were weighed and denoted as m_1 and m_2, respectively. Acid insoluble lignin (Klason lignin) = $(m_2 - m_1)/m_0$ was calculated according to the method described in NREL/TP-510-42618.

The acid soluble lignin (B) in the filtrate was calculated using Eq. (5).

$$B\ (\%) = A/110 \times D \times V \times 100/1000/m_0 \tag{5}$$

where, A is the absorbance of the filtrate, D is the dilution ratio of the filtrate, 110 is the absorption coefficient. V is the total volume of the solution, m_0 is the original mass of the lignin.

Total lignin content is determined as the sum of acid soluble lignin and acid insoluble lignin.

4.13. GC–MS analysis

Lignin oil was dissolved in methanol. Samples were filtered through a 0.22 μm microporous filter membrane, and analyzed using a gas chromatography–mass spectrometry (7890 A-5975C, Agilent Technologies, USA) equipped with a HP-5MS chromatographic column (30.0 m × 250 μm, 0.25 μm). During the analysis, the initial temperature was maintained at 40 °C for 5 min, then increased to 280 °C at a rate of 10 °C/min and held at 280 °C for 10 min.

CRediT authorship contribution statement

Xian-Sheng Cao: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xu-Liang Lin:** Writing – review & editing, Formal analysis. **Bo-Ya Li:** Methodology, Investigation. **Ru-Chun Wu:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Lei Zhong:** Writing – review & editing, Validation, Conceptualization.

Declaration of competing interest

There is no conflict to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2024.130475.

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